SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name: InkPR02NP®

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture: Industrial and research

1.3 Details of the supplier of the safety data sheet

Company	Nanopaint, Lda
Address	IBS – Institute of Science and Innovation for Bio-sustainability Universidade do Minho, Campus de Gualtar 4710-054 Braga, Portugal
Telephone	+351 253 601 681
E-mail address	info@nanopaint-tech.com

1.4 Emergency telephone number

+351 253 601 681

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)



GHS07

Skin Irrit. 2	H315 Causes skin irritation.
Eye Irrit. 2	H319 Causes serious eye irritation.
STOT SE 3	H335 May cause respiratory irritation.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

The product is classified and labelled according to the CLP regulation.

Warning Pictograms:



GHS07

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Hazard Statements:

H315: Causes skin irritation.
H319: Causes serious eye irritation.
H335: May cause respiratory irritation.

Precautionary statements:

P261: Avoid breathing vapours.
P280: Wear protective gloves/protective clothing/eye protection/face protection
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304+P340: IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P403+P233: Store in a well-ventilated place. Keep container tightly closed.
P501: Dispose of contents and/or containers in accordance with regional/national regulations.

2.3 Other hazards

This product does not contain any dangerous ingredients above the limit values.
This product does not contain substances of high concern [Regulation (EC) No 1907/2006 (REACH), Article 1907/2006 57] in an amount ≥ 0.1%.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Chemical nature

1-Isopropyl-4-methylbenzene with carbon additives and auxiliary agents.

Dangerous Components

CAS: 99-87-6 EINECS: 202-796-7	1-Isopropyl-4-methylbenzene	55-80%
	Flam. Liq. 3, H226; Asp. Tox. 1, H304; Skin Irrit. 2, H315; Eye Irrit. 2, H319; STOT SE 3, H335	

Additional information: For the wording of the listed hazard phrases refer to section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

General advice: Get medical attention if irritation or other symptoms occur. Take a copy of the Safety Data Sheet when going for medical treatment.

If inhaled: Remove to fresh air. In case of unconsciousness place patient stably in side position for transportation.

In case of contact with skin: Clean the affected area thoroughly with water and soap. In case of modifications on the skin or discomfort, consult a specialist.

In case of eye contact: Remove contact lenses. Rinse the eyes immediately and thoroughly with water. If eye irritation persists, consult a specialist.

If swallowed: Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

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4.2 Most important symptoms and effects, both acute and delayed

No further relevant information available.

4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media: Use fire extinguishing methods suitable to surrounding conditions.

5.2 Special hazards arising from the substance or mixture

No further relevant information available.

5.3 Advice for firefighters

Special protective equipment for firefighters: It might be necessary to use self-contained respiratory protective device.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Not required.

6.2 Environmental precautions

Do not allow to enter sewers/ surface or ground water.

6.3 Methods and material for containment and cleaning up

Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).
Ensure adequate ventilation.

6.4 Reference to other sections

See Section 7 for information on safe handling.
See Section 8 for information on personal protection equipment.
See Section 13 for disposal information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Ensure good ventilation/exhaustion at the workplace.
Avoid inhalation of spray application of the product.
To remove contaminated clothing and protective equipment before entering eating areas.
To wash hands after use.
Not to eat, drink and smoke in work areas.

7.2 Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

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7.3 Specific end use(s)

Specific use(s): No other specific used except for the ones mentioned in section 1.2.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Ingredients with limit values that require monitoring at the workplace:
The product does not contain any relevant quantities of materials with critical values that have to be monitored at the workplace.

8.2 Exposure controls

Individual protective measures, such as personal protective equipment

General protective and hygienic measures:
Keep away from foodstuffs, beverages and feed.
Immediately remove all soiled and contaminated clothing.
Wash hands before breaks and at the end of work.
Avoid contact with the eyes and skin.

Eye protection: Safety glasses

Hand protection: Protective gloves. The glove material has to be impermeable and resistant to the product/ the substance/ the preparation. Due to missing tests no recommendation to the glove material can be given for the product/ the preparation/ the chemical mixture. Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation

Skin and body protection: Protective clothing. The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

Respiratory protection: Select respiratory protection suitable for the actual or predicted level of exposure, the type of compound and its level of danger, certified in accordance with applicable standards. For low exposure applications, use respiratory masks with adequate protection / filters. For applications with an exposure level above the Workplace Exposure Limits (WEL), use breathing masks with adequate filters or assisted breathing masks, according to the risk assessment carried out by the occupational risk prevention services.

Environmental exposure controls: Prevent further leakage or spillage if safe to do so. Do not let product enter drains.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid
Form	Ink/paste
Colour	Black

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Odour	Characteristic
pH	Not applicable
Melting point/freezing point	No data available
Boiling point or initial boiling point and boiling range	177 °C (CAS: 99-87-6 p-cymene)
Flash point	116 °C (ISO 3679)
Evaporation rate	No data available
Upper/lower explosion limit	Lower: 0.7 Vol % (CAS: 99-87-6 p-cymene) Upper: 5.6 Vol % (CAS: 99-87-6 p-cymene)
Vapour pressure	No data available
Density	0,735 g/cm^3
Water solubility	Insoluble
Viscosity	3000-8000 cP

9.2 Other information

No data available.

SECTION 10: Stability and reactivity

10.1 Reactivity

No further relevant information available.

10.2 Chemical stability

Stable under recommended storage conditions.

10.3 Possibility of hazardous reactions

No dangerous reactions known.

10.4 Conditions to avoid

No further relevant information available.

10.5 Incompatible materials

No further relevant information available.

10.6 Hazardous decomposition products

No decomposition if stored and applied as directed.

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SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity
Based on available data, the classification criteria are not met.

Skin corrosion/irritation
Causes skin irritation.

Serious eye damage/eye irritation
Causes serious eye irritation.

Respiratory or skin sensitisation
Based on available data, the classification criteria are not met.

Germ cell mutagenicity
Based on available data, the classification criteria are not met.

Carcinogenicity
Based on available data, the classification criteria are not met.

Repeated dose toxicity
Based on available data, the classification criteria are not met.

STOT-single exposure
May cause respiratory irritation.

STOT-repeated exposure
Based on available data, the classification criteria are not met.

Aspiration hazard
Based on available data, the classification criteria are not met.

SECTION 12: Ecological information

12.1 Toxicity

No further relevant information available.

12.2 Persistence and degradability

No further relevant information available.

12.3 Bioaccumulative potential

No further relevant information available.

12.4 Mobility in soil

No further relevant information available.

12.5 Results of PBT and vPvB assessment

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

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12.6 Endocrine disrupting properties

The product does not contain substances with endocrine-disrupting properties.

12.7 Other adverse effects

Water hazard class 2 (German Regulation) (Self-assessment): hazardous for water Do not allow product to reach ground water, water course or sewage system. Danger to drinking water if even small quantities leak into the ground.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product

Recommendation: Disposal must be made according to official regulations. Must not be disposed together with household garbage. Do not allow product to reach sewage system. The information given is based on Directive (EU) 2008/98.

European waste catalogue: This product is considered hazardous waste as defined by EU Directive 2008/98/EC. 08 01 11* - waste paint and varnish containing organic solvents or other dangerous substances.

Contaminated packaging

Recommendation: Disposal must be made according to official regulations. The package should be properly drained.

SECTION 14: Transport information

14.1 UN number (ADR, IMDG, IATA)

Not regulated as a dangerous good

14.2 UN proper shipping name

Not regulated as a dangerous good

14.3 Transport hazard class(es)

Not regulated as a dangerous good

14.4 Packing group

Not regulated as a dangerous good

14.5 Environmental hazards

Not regulated as a dangerous good

14.6 Special precautions for user

Not applicable.

14.7 Maritime transport in bulk according to IMO instruments

Not applicable.

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SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

This safety datasheet complies with the requirements of Regulation (EC) No. 1907/2006.

15.2 Chemical safety assessment

A Chemical Safety Assessment has not been carried out.

SECTION 16: Other information

The information contained herein is based on the present state of our knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship. It is the user's responsibility to satisfy himself as to the suitability and completeness of such information for his own particular use. We do not accept liability for any loss or damage that may occur from the use of this information. This SDS provides selected regulatory information on this product, including its components. This is not intended to include all regulations. It is the responsibility of the user to know and comply with all applicable rules, regulations and laws relating to the product being used.

Phrases from sections 2 and 3
H226 Flammable liquid and vapour.
H304 May be fatal if swallowed and enters airways.
H315 Causes skin irritation.
H319 Causes serious eye irritation.
H335 May cause respiratory irritation

Classification according to Regulation (EC) No 1272/2008

Skin corrosion/irritation Serious eye damage/eye irritation Specific organ toxicity (single exposure)	The classification of the mixture is generally based on the calculation method using substance data according to Regulation (EC) No 1272/2008.

Department issuing SDS:
Nanopaint, Lda
IBS – Institute of Science and Innovation for Bio-sustainability

Contact: See address in section 1 on this MSDS

Abbreviations and acronyms:
ADR: Accord relatif au transport international des marchandises dangereuses par route (European Agreement Concerning the International Carriage of Dangerous Goods by Road)
IMDG: International Maritime Code for Dangerous Goods
IATA: International Air Transport Association
GHS: Globally Harmonised System of Classification and Labelling of Chemicals
EINECS: European Inventory of Existing Commercial Chemical Substances
ELINCS: European List of Notified Chemical Substances
CAS: Chemical Abstracts Service (division of the American Chemical Society)
PBT: Persistent, Bioaccumulative and Toxic
SVHC: Substances of Very High Concern
vPvB: very Persistent and very Bioaccumulative
Flam. Liq. 3: Flammable liquids – Category 3
Skin Irrit. 2: Skin corrosion/irritation – Category 2
Eye Irrit. 2: Serious eye damage/eye irritation – Category 2
STOT SE 3: Specific target organ toxicity (single exposure) – Category 3
Asp. Tox. 1: Aspiration hazard – Category 1

For further information please contact **info@nanopaint-tech.com**.

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